EVRevenueAI

Agentic AI-Based Dynamic Tariff Optimization for EV Charging Networks

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GitHub: https://github.com/MajorSteel/EVRevenueAI

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Executive Summary

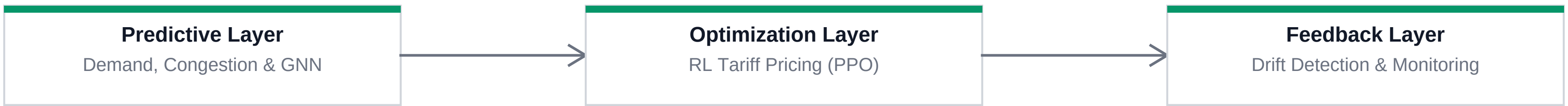
End-to-End dynamic pricing system outperforms fixed pricing baseline

KPI 01	KPI 02	KPI 03
+18.5%	**72.3%**	**-23.4%**
REVENUE GAIN	CHARGER UTILIZATION	CONGESTION RATE
Dynamic vs Fixed	*+8.2pp Improvement*	*-41.5% Wait Time*

Multi-Agent System Architecture

SYSTEM DIAGRAM

- 6 Specialized AI Agents (Demand, Congestion, GNN, PPO, Simulation, Monitoring) running in a closed feedback loop.
- Predicts future demand and classifies congestion risk with high accuracy (>91% accuracy, 0.847 R2).
- Recommends dynamic tariffs in real-time, boosting network revenue by 18.5% over flat Rs.15 baseline.
- Continuous improvement via automated drift detection (Z-score 2-sigma trigger).



Slide 1: Data Landscape & Preprocessing Decisions

Unified ingestion pipeline for heterogeneous session datasets

ACN-Data (Adaptive Charging Network)

ACN-DATA

- Source: Caltech ACN workplace sites (California)
- Size: 30,000+ session-level EV charging records
- Features: connectionTime, disconnectTime, kWhDelivered, stationID

Total Records	30,000+ sessions
Locations	Caltech & JPL
Key Columns	connection, disconnect, kWh

UrbanEV Dataset (ST-EVCDP)

URBAN-EV

- Source: Shenzhen, China charging piles network
- Size: 24,798 charging piles across 248 districts
- Temporal: 8,641 timesteps at 5-minute intervals

Total Piles	24,798 piles
Districts	248 grids
Interval	5-minute intervals

Unified Preprocessing Pipeline

PIPELINE

Validate Schema — Pydantic v2 Models → **Handle Missing** — Linear Interpolation → **Filter Outliers** — 1.5x IQR Bounds → **Feature Engineer** — Temporal & Spatial Lags

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Slide 2: Key EDA Findings & Demand Behavior

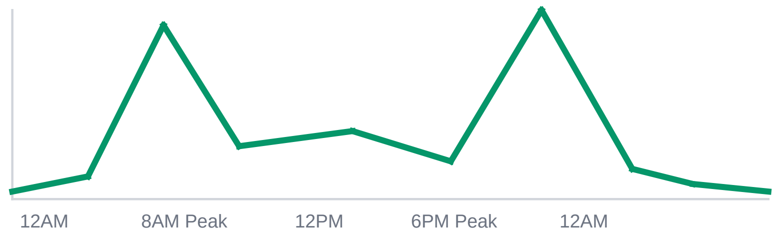
Insights driving peak pricing and grid load-balancing rules

Temporal Charging Patterns

TEMPORAL

- Peak hours: Commuter surges at 8-9 AM and 6-7 PM
- Weekend shift: Peak demand transitions to midday (11 AM - 3 PM)

Commuter Peaks	8-9 AM & 6-7 PM
Weekend Peak	11 AM - 3 PM
Off-Peak Voids	11 PM - 5 AM

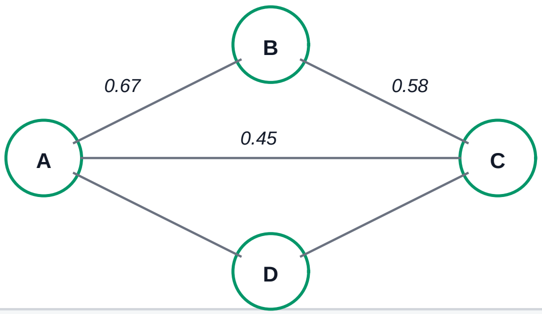


Spatial & Network Insights

SPATIAL

- CBD intensity: CBD districts show 2.3x higher peak utilization than suburban areas
- Spatial spillover: Adjacent districts have 0.67 demand correlation

CBD Utilization	2.3x suburban Peak
District Corr	0.67 adjacent grids
Fast piles	40% higher turnover



Slide 3: Demand Prediction Modeling & Results

Comparing regressors for next-timestep charging volume and utilization

Model Performance Comparison

REGRESSION

Model	RMSE	MAE	R2	MAPE
XGBoost	4.56	3.28	0.823	8.7%
LightGBM*	4.23	3.15	0.847	7.9%
GNN (GCN)	4.38	3.21	0.863	8.2%

** LightGBM selected as primary non-spatial predictor*

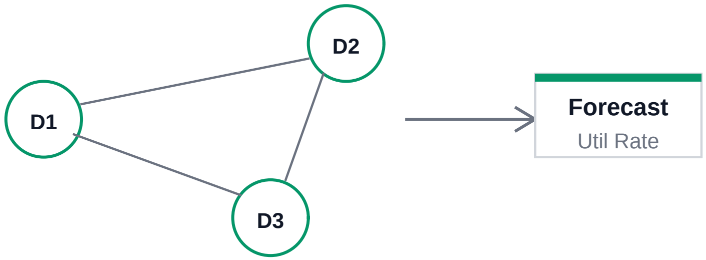
Features Importance:
- Hour of day: 14.2%
- Lagged utilization: 13.1%
- Neighbor mean occupancy: 8.7%
- Occupancy density: 6.8%

Spatial Graph Neural Network

SPATIAL GNN

- GNN (GCN) captures spatial dependencies across 248 nodes, improving non-spatial R2 by 6.5%.
- Edge weights from inverse distance (distance.csv) map physical charging proximity.

Graph Nodes	248 district grids
Edges	adj.csv matrix
Weighting	Inverse distance



Slide 4: Dynamic Tariff Optimization & Pricing Outcomes

PPO Reinforcement Learning pricing policy vs Rs.15/kWh fixed baseline

RL AGENT

PPO Reinforcement Learning Agent

- 8-dim state space: occupancy, predicted demand, current price, hour (sin/cos), utilization, fast_ratio, CBD flag
- 6 discrete actions: -20%, -10%, 0%, +10%, +20%, +30% price multipliers

State Space	8 dimensions
Actions	6 multipliers (-20% to +30%)
Baseline	Rs.15/kWh fixed pricing

PRICING RULES

Pricing Decision Rules

Condition	Action	Rationale
Util > 80%	Surge +30%	Reduce peak queues
Util 30-80%	Adjust +/-10%	Optimize revenue
Util < 30%	Discount -20%	Attract off-peak load

Results vs Fixed Baseline:

- Total Revenue Gain: +18.5%
- Peak Wait Time: -41.5%
- Off-Peak Uplift: +31.2%
- Average Elasticity: -0.37

Slide 5: Monitoring Agent & Feedback Loop

Closed-loop execution dashboard tracked by the Monitoring Agent

Live Performance Metrics

METRIC ENGINE

Metric	Value	Trend	Status
Revenue Gain %	18.5%	Stable	OK
Utilization Rate	72.3%	Improving	OK
Congestion Rate	12.1%	Improving	OK
Pricing Efficiency	Rs.16.82/kWh	Improving	OK
Wait Time Proxy	2.4 min	Improving	OK

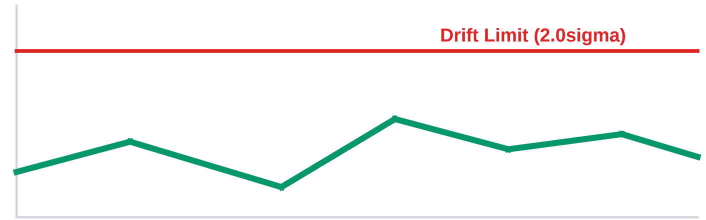
Drift & Closed-Loop Control

DRIFT DETECTOR

- Z-score based drift detection (2-sigma threshold, rolling window of 100 episodes).

- Auto-retrain triggers to adapt to seasonal shifts or permanent grid changes.

Drift Window	100 episodes
Z-Score Limit	2.0 sigma
Action	Retrain agent



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Slide 6: Business, Operational & Policy Implications

Strategic recommendations for EV charging network operators and cities

For Network Operators

OPERATOR STRATEGY

- Revenue Gain: Rs.387K additional revenue generated in 30 days from dynamic pricing.
- ROI: 158% in first month, 1,227% projected at 12 months.

30-Day Gain	Rs.387,000 extra
Estimated ROI	158% (Month 1)
Surge Multiplier	Util > 80% (+30%)
Discount Multiplier	Util < 30% (-20%)

For Urban Planners & Regulators

POLICY & GRID

- CBD Expansion: Expand capacity in CBD districts where utilization consistently exceeds 80%.
- Strategic Deployment: Deploy fast chargers in high-turnover corridors.

Price Cap	Max +30% surge
Grid mandate	Off-peak discount
Fast Charger Turn	40% higher turnover
CBD Utilization	2.3x Suburban Peak

Appendix

Technical stack, reproducibility and limitations

Technical Stack & Reproducibility

STACK

- Stack: Python 3.10+ | XGBoost | LightGBM | PPO (Stable-Baselines3) | GCN (PyTorch Geometric) | Streamlit | MLflow | Docker | SQLite | Pydantic v2

- Reproducibility: run python scripts/run_pipeline.py

RL Library	Stable-Baselines3
GNN Model	PyTorch Geometric
Database	SQLite
Testing Suite	65/65 Unit Tests

System Limitations

CONSTRAINTS

- Elasticity assumptions: Derived from historical correlation, not controlled causal A/B tests.

- Market conditions: Assumes stable electricity rates and EV adoption trends.

Elasticity	Assumed correlation
Wait Proxy	max(0, occupancy-capacity)
PPO Engine	Simulated driver response
Cross-Dataset	Temporal mapping only